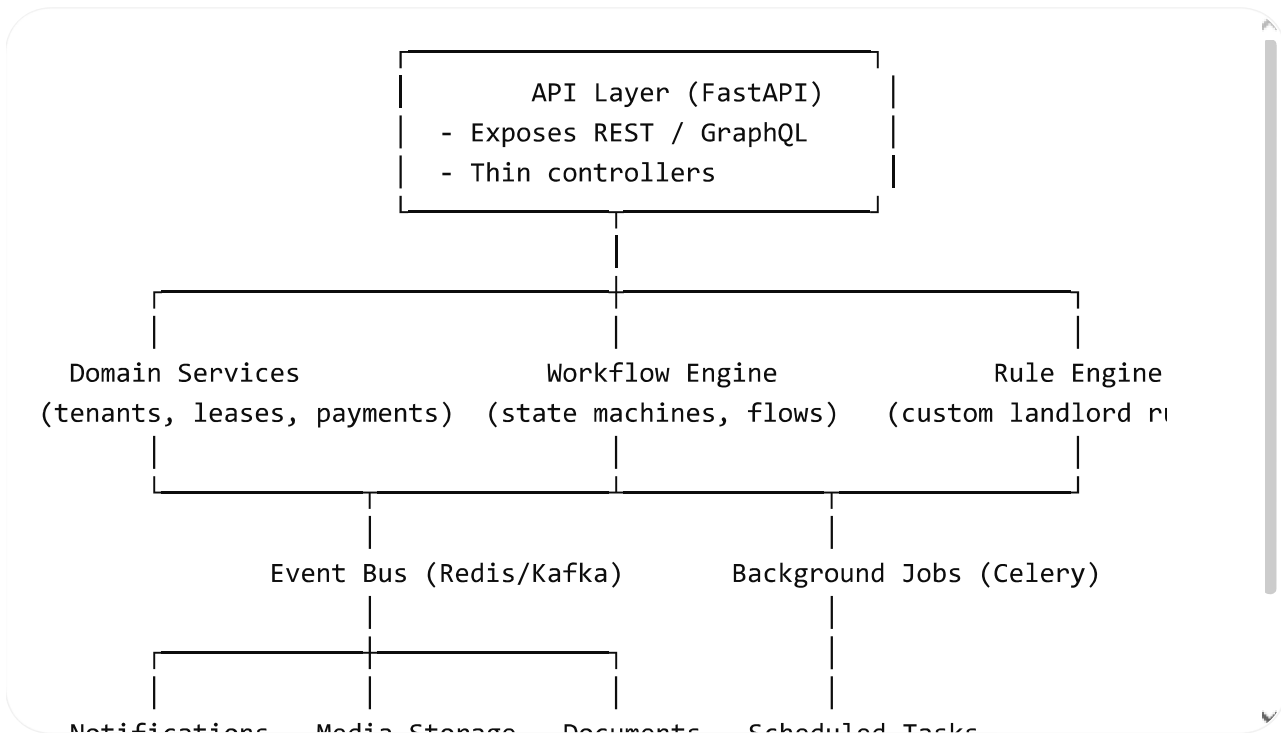


1. Architecture Overview



2. Multi-Tenant DB Strategy (Hybrid)

- **Small landlords / <50 units:** Shared DB → `tenant_id` column for isolation
- **Large landlords / hostels / 200+ rooms:** Dedicated DB
- **Migration tooling** to automate schema updates across DBs
- **Central meta DB:** Tracks landlords, tenants, and global analytics

3. Workflows & State Machines

Flow	Key States	Key Events
Tenant Onboarding	Invited → Started → Submitted → Approved → Activated	INVITE_SENT, ONBOARDING_COMPLETED, TENANT_APPROVED
Lease Lifecycle	Draft → Pending → Active → Notice → Closed	LEASE_CREATED, LEASE_ACTIVATED, NOTICE_GIVEN, LEASE_TERMINATED
Rent / Billing	Scheduled → Due → Overdue → Settled	RENT_GENERATED, RENT_DUE, RENT_OVERDUE
Payment	Initiated → Pending → Completed → Failed →	PAYMENT_RECORDED, PAYMENT_CONFIRMED,

Flow	Key States	Key Events
	Reconciled	PAYMENT_FAILED
Late Fee	Eligible → Applied → Paid	LATE_FEE_TRIGGERED, LATE_FEE_APPLIED
Inspection	Scheduled → In Progress → Completed → Approved	INSPECTION_CREATED, INSPECTION_COMPLETED
Vacating	Notice Given → Scheduled → Inspected → Settled → Closed	NOTICE_SUBMITTED, MOVE_OUT_SCHEDULED, DEPOSIT_SETTLED
Document	Generated → Sent → Viewed → Accepted → Archived	DOCUMENT_GENERATED, DOCUMENT_ACCEPTED
Notification	Queued → Sent → Delivered → Read → Failed	MESSAGE_SENT, MESSAGE_FAILED
Maintenance	Reported → Assigned → In Progress → Resolved → Closed	ISSUE_CREATED, ISSUE_RESOLVED

4. Integration & Communication Layer

- **WhatsApp / SMS / Email:** Abstracted provider layer, queue-based, deduplicated
- **Document Generation:** PDFs + e-sign, cloud storage (S3)
- **Media Storage:** Inspection photos/videos, compressed + tiered storage
- **Payments:** Bank transfer reconciliation + optional mobile money / online payments

5. Tasks & Background Jobs

- Celery workers handle:
 - Rent reminders & overdue notifications
 - Late fee calculation & application
 - Deposit refunds processing
 - Tenant inactivity / onboarding reminders
- Supports **horizontal scaling** for large hostels / multi-room properties

6. Rule Engine & Customization

- **Per landlord property rules:** stored as JSON or DSL
 - **Dynamic evaluation:**
 - Rent cycles
 - Deposit amounts
 - Late fees and grace periods
 - Notice periods
 - Validation engine prevents conflicting rules
-

7. Audit & Compliance

- Immutable logs of all workflow transitions
 - Payment & deposit history
 - Document access logs
 - GDPR-ready: tenant data deletion & anonymization on request
-

8. Suggested Folder Structure (Implementation-Ready)

```
app/
├── api/
│   └── v1/                # Thin routers per module
├── core/                  # Config, security, middleware, logging
├── db/                    # DB setup & base models
├── events/                # Event bus & handlers
├── integrations/
│   ├── whatsapp/
│   ├── storage/
│   └── documents/
├── modules/
│   ├── tenants/
│   ├── leases/
│   ├── payments/
│   ├── inspections/
│   ├── notifications/
│   └── properties/
└── tasks/                 # Celery workers & scheduled jobs
```

9. Key Advantages of This Architecture

- **Future-proof:** Can scale from 5 units → 10,000 units without major rewrites
- **Tenant onboarding frictionless:** Links, forms, docs handled asynchronously
- **Highly customizable:** Each landlord can define their own rules

- **Resilient:** Queued events + deduplicated notifications handle outages
- **Audit-ready:** Complete logs for disputes, finance, and GDPR